

Business Summary Report: Predictive Insights for Collections Strategy

1. Summary of Predictive Insights

- Customers with higher Credit Utilization show slightly higher delinquency rates, with an average of 0.51 compared with 0.49 for non-delinquent customers. This suggests that high credit usage could be a useful early warning sign.
- Missed or late payments and a higher Debt-to-Income Ratio also appear relevant to delinquency risk. Delinquent customers had a slightly higher average DTI (0.31 vs. 0.30).
- Some customer groups show higher observed delinquency rates, particularly unemployed customers (19.4%), Business card holders (21.3%), and Student card holders (17.9%). These patterns should be monitored, but they should not be treated as definite causes of delinquency without further validation.

Key Insight	Customer Segment	Influencing Variables	Potential Impact
Higher credit utilization is associated with slightly higher delinquency risk	Customers with higher credit utilization	Credit Utilization, Missed Payments, Debt-to-Income Ratio	Prioritize proactive payment reminders and early outreach.
Unemployed customers show a higher observed delinquency rate.	Unemployed customers	Employment Status, Payment History	Consider supportive and targeted outreach for customers showing additional risk signals.
Business and Student cardholders show higher observed delinquency rates.	Business and Student cardholders	Credit Card Type, Payment History	Monitor these segments more closely and assess whether additional risk factors are present.

2. Recommendation Framework

- **Restated Insight:** Customers with higher Credit Utilization show slightly higher delinquency risk (0.51 vs. 0.49).
- **Proposed Recommendation:** Pilot targeted early-payment reminders for customers with high Credit Utilization.
 - **Specific:** Send proactive payment reminders to identified high-utilization customers before their payment due dates.

- **Measurable:** Aim for a 10% reduction in late or missed payments among the targeted customers during the pilot.
- **Actionable:** Use the existing Collections outreach process to contact customers identified as higher risk.
- **Relevant:** This allows Geldium to intervene earlier and potentially prevent customers from becoming delinquent.
- **Time-bound:** Run the pilot for 6 weeks and review the results against the current baseline.
- **Justification and Business Rationale:** A reminder campaign is relatively simple to implement, can be scaled if successful, and is less intrusive than stronger collection actions. Measuring the results during a short pilot will help Geldium determine whether the approach actually reduces late or missed payments before expanding it.

3. Ethical and Responsible AI Considerations

The model and recommendation should be used carefully to make sure customers are treated fairly and that AI supports, rather than replaces, human judgment.

- **Fairness and bias:** The model should be checked to make sure it does not unfairly identify certain customer groups as high risk. For example, Employment Status and Location should be reviewed to see whether prediction patterns differ significantly across groups. Location should also be checked for possible proxy bias.
- **Explainability:** Logistic Regression is relatively easy to explain because the factors influencing a prediction can be clearly communicated. For example, the Collections team can explain that a customer was flagged because their credit utilization and payment history show patterns associated with higher delinquency risk.
- **Responsible decision-making:** The recommended reminder campaign is a relatively supportive and low-impact intervention. The model should be used to prioritize outreach rather than automatically determine how customers are treated, with human review remaining part of the process.
- **Transparency and accountability:** Geldium should clearly communicate how the model is being used, regularly review its performance and fairness, and investigate unexpected outcomes. Customer information should also be handled responsibly and kept secure.